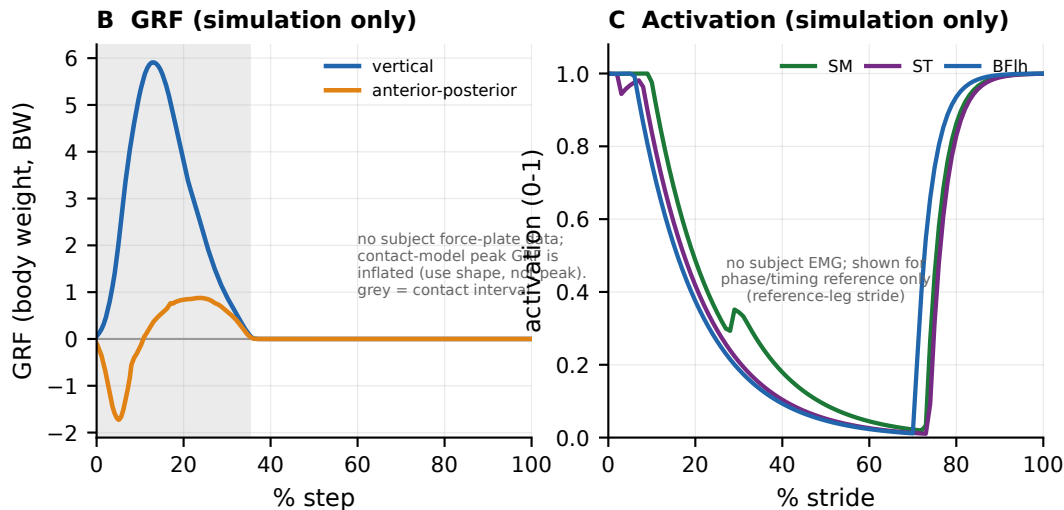
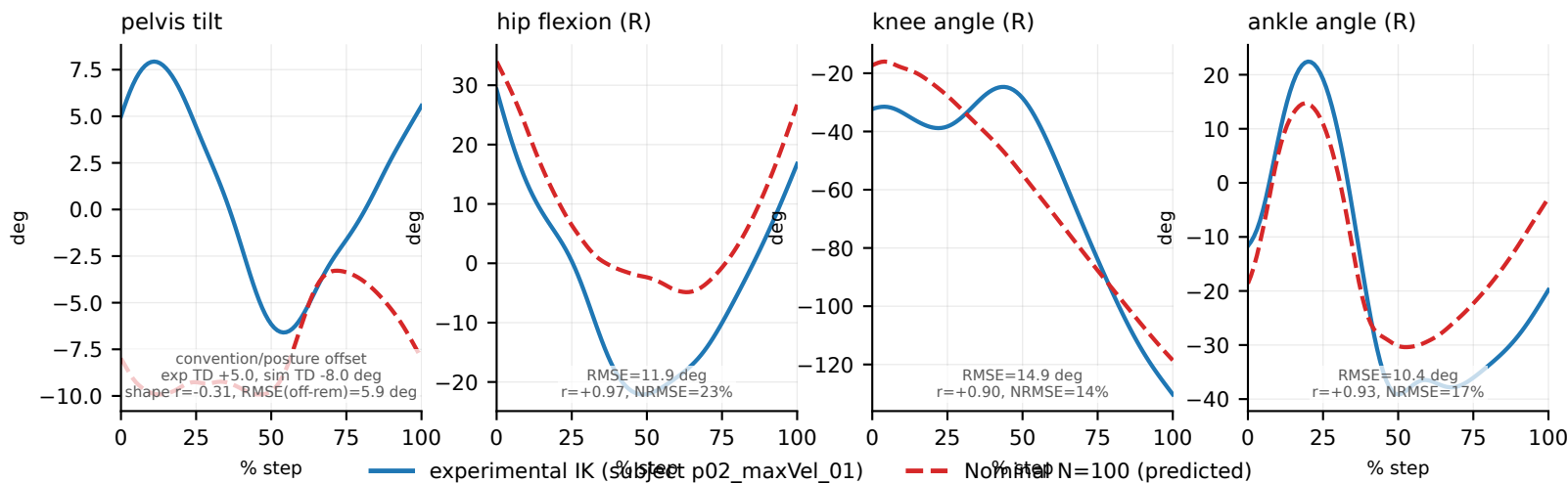


Figure 4 | Baseline (Nominal N=100) reproduction of the subject's own max-velocity sprint step



D Quantitative agreement

kinematics (exp IK vs Nominal N=100):

	RMSE(shape)	RMSE	NRMSE	r
pelvis tilt	5.9 deg			$r=-0.31$ *
hip flexion (R)	11.9 deg	23%		$r=+0.97$
knee angle (R)	14.9 deg	14%		$r=+0.90$
ankle angle (R)	10.4 deg	17%		$r=+0.93$

peak vertical GRF (sim) = 5.9 BW
GRF timing error : not available
EMG onset/offset : not available

* pelvis_tilt convention/posture offset removed for shape comparison

External kinematic / GRF-shape / activation-timing agreement does NOT guarantee validity of internal fibre length, force or local tissue stress. GRF and activations are simulation-only (no subject force plate or EMG in the dataset).